A PROJECT ON

PREDICTIVE ANALYSIS OF CUSTOMER CHURN AND RETENTION STRATEGY

USING IBM SPSS MODELER

FACULTY OF COMMERCE AND MANAGEMENT

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SUBMITTED BY:

ADITYA PRATAP SINGH

BBA in collaboration with IBM

Roll No. 24051020002

SUPERVISED BY:

Mr. FERHAN KHAN

UNITED UNIVERSITY

RAWATPUR, PRAYAGRAJ, UTTAR PRADESH – 211012

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1. INTRODUCTION

Customer retention is an important factor in the long-term success of service businesses. Acquiring a new customer often requires marketing expenditure, sales effort and onboarding support. When existing customers discontinue a service, the organization may lose recurring revenue and incur additional costs to replace them. Businesses therefore use data analytics to identify customers who may be at risk of leaving.

Customer churn refers to a customer stopping the use of a company's product or service during a defined period. Churn prediction applies statistical and machine-learning techniques to historical customer information to estimate the likelihood that a customer will leave. The results help managers prioritize retention activities instead of applying the same campaign to every customer.

IBM SPSS Modeler is a visual data-mining and predictive-analytics platform. Users build analytical streams by connecting source, preparation, modeling and output nodes. This project presents an academic case study of ConnectPlus Telecom, a fictional subscription-based provider. A simulated dataset of 5,000 customer records demonstrates data preparation, exploratory analysis, classification modeling and customer-risk interpretation.

The project examines customer tenure, monthly charges, contract type, service usage, support interactions and payment method. All sample statistics and model results in this report are illustrative learning examples, not actual company performance.

2. OBJECTIVES OF THE STUDY

- To understand customer churn and its importance for subscription-based businesses.
- To prepare a structured customer dataset for IBM SPSS Modeler.
- To compare customer characteristics across churn and non-churn groups.
- To identify variables associated with a higher likelihood of churn.
- To build a classification model that estimates churn probability.
- To evaluate performance using accuracy, precision, recall and ROC-AUC.
- To translate model outputs into practical customer-retention recommendations.

3. LITERATURE REVIEW / THEORETICAL BACKGROUND

3.1 Importance of Customer Retention

Customer churn is the loss of customers over a specified period. In telecom, banking, insurance, streaming and other subscription businesses, churn can reduce recurring income and weaken customer relationships. Monitoring churn helps organizations understand whether customers are dissatisfied with price, service quality, product value or support.

3.2 Predictive Analytics in Customer Retention

Predictive analytics uses historical observations to estimate future outcomes. In churn analysis, the outcome is commonly represented as a binary field: Churn = Yes or Churn = No. A model learns patterns from customer attributes and uses those patterns to score customers whose future behavior is not yet known.

3.3 Classification Techniques

Classification is supervised learning used when the target variable represents a category. Common techniques include logistic regression, decision trees, neural networks, random forests and gradient boosting. Decision trees are relatively easy to interpret, while logistic regression provides probability-based estimates. The appropriate model depends on data quality and business requirements.

3.4 Model Evaluation

Accuracy measures the proportion of all predictions that are correct. Precision measures how many customers predicted as churners actually churn. Recall measures how many actual churners are identified. The F1-score combines precision and recall. ROC-AUC evaluates how well a model ranks positive cases above negative cases across different thresholds.

3.5 IBM SPSS Modeler

SPSS Modeler provides a visual stream environment for importing data, setting field types, handling missing values, exploring distributions, building models and reviewing output. Its modeling palette includes classification methods such as C5.0, CHAID, logistic regression and neural networks. The workflow can be refreshed when new customer records become available.

4. RESEARCH METHODOLOGY

The project follows a quantitative and applied analytical approach. The workflow is designed around a simulated customer dataset and an end-to-end stream in IBM SPSS Modeler:

- Problem Definition – define churn as the customer outcome to be predicted.
- Data Preparation – create a structured sample of 5,000 customer records in spreadsheet format.
- Data Understanding – inspect field types, missing values, unusual values and category distributions.
- Exploratory Analysis – compare churn across customer groups.
- Model Development – train classification models using selected customer variables.
- Model Evaluation – assess performance on a holdout sample.
- Interpretation – identify high-risk segments and suggest retention actions.

Software Used: Microsoft Excel for preparing the sample dataset and IBM SPSS Modeler for data auditing, visualization, classification modeling and scoring.

5. DATA DESCRIPTION AND PREPARATION

5.1 About the Dataset

The illustrative dataset contains 5,000 customer records for ConnectPlus Telecom. Each row represents one customer at an observation date. The target field is Churn, which records whether the customer discontinued the service during the defined period.

Variable	Description	Type / Unit
Customer_ID	Unique customer reference	Identifier
Tenure_Months	Length of relationship	Integer / months
Contract_Type	Monthly, one-year or two-year	Categorical
Monthly_Charges	Recurring monthly bill	Continuous
Data_Usage_GB	Average monthly data use	Continuous / GB
Support_Calls	Support calls in period	Integer / count
Payment_Method	Customer payment method	Categorical
Auto_Pay	Automatic payment enabled	Yes / No
Late_Payments	Number of late payments	Integer / count
Churn	Whether customer left	Target / Yes-No

5.2 Data Characteristics

The sample includes numerical and categorical fields. Tenure and contract duration describe the length and stability of the relationship. Monthly charges and data usage represent service and spending patterns. Support calls and late payments may indicate friction in the customer experience. Payment method and automatic payment status provide additional behavioral information.

5.3 Data Preparation Steps

- Check that Customer_ID is unique and Churn contains only Yes or No.
- Convert tenure, charges, data usage, support calls and late payments to numeric measurement levels.
- Set contract type, payment method, Auto_Pay and Churn as categorical fields.
- Review missing values and handle them according to their meaning.
- Check for impossible values, such as negative tenure or charges.
- Keep Customer_ID as an identifier and do not use it as a predictive input.
- Partition records into training and testing samples.

A consistent preparation process is essential because incorrect field roles or inconsistent categories can affect model training and final scores.

6. BUILDING THE SPSS MODELER STREAM

6.1 Recommended Stream Structure

Excel Source → Type → Data Audit → Select / Derive → Graphs → Partition → Classification Model → Analysis / Evaluation → Table / Export

Detailed node sequence:

- Excel Source node – import the customer dataset and confirm field names.
- Type node – assign measurement levels and define Churn as the target.
- Data Audit node – inspect distributions, missing values, outliers and field quality.
- Select and Derive nodes – remove identifier fields from model inputs and create required fields.
- Graphboard / Distribution nodes – visualize churn by contract type, tenure and payment category.
- Partition node – create training and testing samples.
- Modeling node – train a decision tree or logistic regression model.
- Analysis / Evaluation nodes – review predictions and classification quality.
- Table / Export nodes – save scored records and risk categories.

6.2 Key Settings in Important Nodes

Type node: set Churn as the categorical target. Set numerical measures to continuous and service/payment descriptors to categorical. Customer_ID should be an identifier or have no modeling role.

Partition node: use training records for model development and a separate testing sample for independent evaluation. An illustrative split is 70% training and 30% testing. Use a reproducible random seed where available.

Classification node: choose Churn as the target and use relevant predictors such as tenure, contract type, monthly charges, support calls, payment method, automatic payment and late payments. Exclude Customer_ID.

7. DESCRIPTIVE STATISTICS AND EXPLORATORY ANALYSIS

Exploratory analysis helps understand the dataset before modeling. The following summary values are illustrative characteristics of the simulated dataset, not output from a live SPSS execution.

Statistic	Tenure (months)	Monthly Charges	Data Usage (GB)	Support Calls
N	5,000	5,000	5,000	5,000
Mean	28.4	74.6	18.2	1.8
Std. Deviation	17.6	22.1	9.4	1.5
Minimum	1	20.0	0.5	0
Maximum	72	150.0	60.0	12

Interpretation: customers are at different stages of their relationship with the provider. Monthly charges and data usage vary considerably, while support-call counts show that some customers contact the service team more frequently.

7.1 Exploratory Comparisons

Useful visualizations include churn by contract type, a histogram of customer tenure, a box plot of monthly charges by churn status and average support calls by churn status. These views can reveal patterns to investigate further.

7.2 Illustrative Churn Distribution

Customer Status	Records	Share
Churn = No	3,650	73%
Churn = Yes	1,350	27%
Total	5,000	100%

The sample is intentionally imbalanced, with fewer churn cases than retained customers. This makes it important to examine recall and precision in addition to overall accuracy. A model that predicts every customer as retained could achieve superficially high accuracy while failing to identify churners.

7.3 Patterns to Examine

- Contract type – compare churn rates for monthly, one-year and two-year contracts.
- Tenure – check whether customers in their first months show different churn behavior.
- Support calls – examine whether repeated service contacts are associated with churn.
- Late payments – compare churn across payment-behavior groups.
- Monthly charges – check whether higher charges coincide with churn while considering usage and contract type.

These comparisons are descriptive and do not prove that any factor causes churn.

8. CLASSIFICATION MODELING AND MODEL EVALUATION

8.1 Model Development

A classification model is developed with Churn as the target. Training records are used to learn relationships between customer attributes and churn status. A decision tree is suitable for an academic demonstration because its rules can be communicated clearly. Logistic regression may also be developed as a benchmark.

8.2 Illustrative Model Evaluation

Metric	Illustrative Result	Meaning
Accuracy	82%	Share of test records classified correctly
Precision (Churn)	68%	Share of predicted churners who actually churn
Recall (Churn)	71%	Share of actual churners identified
F1-score (Churn)	69%	Combined measure of precision and recall
ROC-AUC	0.84	Ability to rank churners above non-churners

Note: these values are example figures created to explain interpretation. They are not verified results from running a dataset in SPSS Modeler.

8.3 Interpretation of Evaluation Measures

An illustrative accuracy of 82% suggests that most test records are classified correctly. Churn precision of 68% means that approximately two-thirds of customers flagged by the model are actual churners. Recall of 71% means that the model identifies about seven out of ten customers who churn in the test sample. The F1-score summarizes the balance between precision and recall.

A business should select the classification threshold according to the cost of retention offers and the financial impact of losing a customer. A lower threshold may identify more possible churners but can also create more false alarms. A higher threshold may reduce unnecessary offers but miss customers who need attention.

9. CUSTOMER CHURN PREDICTION WITH SPSS MODELER

9.1 Scoring New Customer Records

After selecting and evaluating a model, the model nugget can be applied to new customer records. The scored output may contain a predicted class and a probability or confidence score for each class. The churn probability can be used to create risk bands for operational follow-up.

9.2 Suggested Risk Categories

Predicted Churn Probability	Risk Category	Suggested Interpretation
Below 0.30	Low	Routine service communication
0.30 to 0.59	Medium	Review account activity and service concerns
0.60 and above	High	Prioritize personalized retention contact

These probability bands are proposed for demonstration only. A real organization should calibrate thresholds using validation results, retention capacity, customer value and intervention cost.

9.3 Example Customer Scoring Workflow

- Import the latest customer file using a Source node.
- Apply the same field definitions and preparation rules used during model development.
- Connect prepared records to the selected model nugget.
- Run the stream and inspect predicted churn class and probability fields.
- Sort records by predicted churn probability and assign operational risk categories.
- Export scored output for authorized retention teams, subject to privacy and governance controls.

10. PREDICTED CHURN RISK AND RETENTION PRIORITIES

The model output can help a business decide which customer groups should receive attention first. The following sample distribution is an illustrative planning example based on 5,000 customers; it is not a live model output.

Risk Segment	Customers	Share	Suggested Priority
High risk	750	15%	Immediate personalized outreach
Medium risk	1,250	25%	Targeted service review or relevant offer
Low risk	3,000	60%	Regular engagement and monitoring
Total	5,000	100%	—

10.1 Retention Actions by Segment

High-risk customers: review recent complaints, billing concerns and usage changes. Where appropriate, assign a trained service representative to understand the customer's concern.

Medium-risk customers: use relevant education, service check-ins or plan reviews. Offers should be based on customer needs rather than sent automatically to every account.

Low-risk customers: maintain service quality, provide clear communication and recognize loyalty. A low predicted risk does not mean that the customer can be ignored.

10.2 Monitoring Results

The retention team should record whether a customer was contacted, what concern was identified, what action was offered and whether the customer remained active. This feedback can help evaluate campaign effectiveness and improve future model development.

11. FINDINGS AND DISCUSSION

- Customer churn is important because it can reduce recurring revenue and increase customer-acquisition costs.
- The illustrative dataset shows why tenure, contract type, charges, service interactions and payment behavior are useful fields to examine.
- Exploratory analysis helps identify differences between churned and retained customers before model building.
- Classification models can convert historical patterns into churn probabilities that support prioritization.
- Accuracy alone is not sufficient; precision, recall, F1-score and ROC-AUC provide additional information.
- Risk bands can help retention teams organize work when thresholds align with business capacity and intervention costs.
- Predictions should support human decisions and be reviewed alongside customer feedback and service records.

Discussion

The project illustrates the connection between data analytics and customer relationship management. A churn score is not itself a retention strategy. Its value comes from combining the score with timely, relevant and respectful customer support. For example, a customer with repeated support calls may need a service-resolution intervention rather than a generic discount.

The sample evaluation values show how a classification report may be interpreted, but actual business use requires execution on real, appropriately governed data. Model performance should be monitored because customer preferences, prices, competitors and service conditions change.

12. RECOMMENDATIONS

- Early-tenure engagement: provide onboarding, product guidance and service information during the first months.
- Service recovery: review repeated support contacts and resolve the underlying issue rather than relying only on promotional offers.
- Contract education: explain the value, terms and flexibility of longer-term plans without pressuring customers.
- Billing transparency: make charges, payment dates and plan changes easy to understand; provide timely reminders.
- Personalized retention: use customer needs and service history to guide offers instead of sending identical promotions.
- Model refresh: retrain and validate periodically with recent customer data and track performance changes.
- Threshold review: consider false positives, missed churners, intervention costs and expected customer value.
- Data privacy: restrict access to customer-level predictions and follow applicable privacy requirements.
- Campaign measurement: compare retention outcomes for contacted and comparable non-contacted customers.

Implementation should begin with a small pilot. The organization can test the workflow with a limited customer segment, gather feedback from service teams and revise the model and retention process before wider deployment.

13. LIMITATIONS OF THE STUDY

The dataset used in this academic project is simulated for learning purposes. It does not represent actual customer records from a telecom company, and the illustrative statistics and model-evaluation values should not be interpreted as verified business findings.

Model performance depends on the quality, completeness and relevance of input data. If important factors such as network reliability, competitor offers, customer complaints or changes in household circumstances are unavailable, the model may not capture the full reasons behind churn.

The relationship between a predictor and churn does not automatically establish causation. Repeated support calls may be associated with churn because customers are experiencing problems, but the calls themselves may not cause the customer to leave.

Class imbalance can affect model training and evaluation. Accuracy may hide poor identification of the churn class, so several metrics and a suitable validation strategy are required. Probability scores may also need calibration before business use.

Customer behavior changes over time. A model trained on older data may become less accurate when prices, service plans, market conditions or expectations change. Periodic monitoring and retraining are therefore necessary.

Customer-level predictions involve personal data. Any practical implementation must follow privacy, security, access-control and data-retention requirements.

14. CONCLUSION

This project demonstrates an end-to-end academic workflow for customer churn prediction using IBM SPSS Modeler. It explains how a structured customer dataset can be imported, checked, explored and used to develop a classification model. It also shows how predicted churn probabilities can be grouped into risk categories and connected to practical retention actions.

Predictive analytics can help organizations move from reactive customer management toward earlier identification of possible churn. However, a prediction is not a certainty or a substitute for customer understanding. Reliable data, appropriate model evaluation, ethical use and effective follow-up are necessary for a useful churn-management process.

The workflow can be adapted to real organizational data after validation, privacy review and testing. When used responsibly, IBM SPSS Modeler can support data-informed decisions that improve customer experience and strengthen long-term relationships.